Vijay Arunkumar

New York, NY

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https://thekumar.github.io

AI

VertexAI, OpenAI, Langchain, RAG, Knowledge Graphs, PyTorch, ONNX

Web Front-end

React, Redux, Typescript, Javascript, GraphQL, HTML5, Google Closure, Service Workers, CSS.

Web Back-end

Node.js, Express, Apollo GraphQL, Prisma ORM, Java SpringBoot, GoMux, GORM, SQL (MySQL/ Postgres), NoSQL (HBase/ DynamoDB/Mongo).

Cloud & Distributed Systems

Vercel, Firebase, AWS, GCP, Docker, Kubernetes, Terraform, REST, gRPC, Operational Transforms, CRDTs.

Areas of Expertise

Service Oriented Architecture, SQL & NoSQL, LLMs, Deep Learning and Neural Nets, Map Reduce, Search, Recommender Systems and Data Mining.

Languages

Java, Go, Typescript, Javascript, Objective-C, Swift, C++, Python

Published iOS Apps



Fotoyaki
ftyki.com



ZDO
zdoapp.com

Profile

Software Engineering Lead specializing in full-stack product development and scaling systems from zero to 10^n. Curious about the cutting edge. Obsessed about keeping it simple.

Experience

Software Engineer, Meta — Menlo Park, CA / New York, NY 10/24 - now

Member of the Poly-source Sparse Machine Learning XFN Infra team — Privacy Preserving ML and Opt-out Signal Growth in Ads & Monetization Infra.

- Designed and Implemented Differential Privacy techniques in our Ads Ranking model training (Randomized Response, RAPPOR, Label Flipping, etc.), and introduced methods to extract signal from noisy privacy preserving proxy signals across the model stack (Noise Aware Gradient Adaptation, Label Importance, Disagreement Regularization, etc.) unlocking > 1% revenue gain.
- Built agentic flows for accelerated tech harvesting, and for data privacy governance/enforcement.

Software Engineer, Variance (Intrinsic) — San Francisco, CA 06/24-10/24

Core member of 5-person founding team scaling content moderation workflows with Applied AI.

- Owned RedBubble account end-to-end from pilot (5 policies) to productionization (5000 policies).
- Prompt Engineering, Caching, Fine-tuning, Backtesting, Policy Authoring UX, the works!
- Implemented a gated multi-stage indexing and evaluation pipeline to cut down our token usage costs by 10,000x from an estimated $22M/mo to $3000/mo while improving recall and precision.

Founder, Yaki, Inc. — Palo Alto, CA 04/23-06/24

Iterated towards product market fit with LLMs.

- Datayaki.com, a privacy-first data analytics and sharing platform for numerical analysis using LLMs.
- Sailfin.ai, a financial tool for analyzing earnings statements and news about public companies.
- Open sourced @yaki-inc/crypto type safe encryption library with support for encrypted permissions and authorization flows.

Principal Engineer, Skiff — New York, NY / San Francisco, CA 04/22-01/23

Oversaw architectural decisions in addition to full-stack feature development.

- Led effort to improve our service's reliability and performance to handle 10x growth in users.
  - Improved GraphQL API p95 response time by 20x from 1.2s to 65ms.
  - Simplified the product UX and DB schema to scale RDS instance perf by ~100x.
- Designed and implemented web3 wallet-based addressing for Skiff Mail and integrated with ENS.
- Consolidated GraphQL APIs to streamline app development between Pages, Mail, and Calendar.

Software Development Engineer, AWS Billing, amazon — New York, NY 04/20-4/22

Technical Member of Enterprise Support Billing team within the AWS Commerce Platform group.

- Designed an automated customer onboarding workflow employing SNS, CloudTrail, and AWS Lambda to cut lead time from ~2 weeks to ~2 hours.
- Improved speed, accuracy, and reliability of our monthly bill-runs through better parallelization scheduling and chunked file processing to avoid OOM issues.
- Improved code deployments and remote debugging procedures for customers on GovCloud..

Senior Software Engineer, Dropbox — New York, NY 07/18-02/20

- Delivered HIPAA Compliance for Paper. Worked closely with legal and PM to get this done.
- Designed and implemented layout editor and widgets for our Showcase product.
- Improved code reliability of Showcase through comprehensive unit and integration test coverage.
- Analyzed and streamlined our Bazel dependency graph to vastly improve build and CI/CD times.
- Improved devbox memory footprint via a shared service sandbox and a sidecar proxy to intelligently offload calls to sandbox.

Senior Software Engineer, Google — New York, NY 01/16-12/17

Senior member of the Docs Comments team.

- Implemented Instant Mentions (plus/at mention) across all platforms (Javascript, Android, and iOS).
- Primary POC for our iOS code for reviewing external teams' features, designs, and code.
- Designed and implemented "Action Items" for iOS (Objective-C and Javascript).
- Led the team to improve Comments UI performance resulting in > 98% savings in render time.

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Data Science Consultant, DeepDyve — Redwood City, CA 06/15-12/15

- Built the "Others also viewed" and "More from the authors" search recommendation features to rank related articles based on link analysis of sequential views, and shared authors.
- Owned end-to-end design and delivery from tracking web views to ranking suggestions at scale using Hadoop MR.

Senior Web Engineer, Shutterfly — Redwood City, CA 06/13-12/15

As part of the Shared Services Team, rearchitected our monolithic codebase to be more horizontally scalable using RESTful micro services built atop Spring Boot, and Netflix's OSS stack.

- Designed and implemented Single Sign On and OAuth for API access between our brands.
- Designed and implemented the Store Catalog and Product Pricing services using Spring Boot.
- Led the effort around secure design practices and introduced STRIDE modeling to Eng teams.

Engineer, Bread Labs — San Francisco, CA 03/12-05/13

Member of a four-person engineering team working on our Ad Platform (Groovy & Grails on Heroku).

- Designed and implemented our product's subscriptions and payments system with support for pro-rating, invoicing, refunds, discounts, and payment processing using Braintree Payments.
- Designed and built geo-targeting for our ads.
- Also built: Campaign Pacing, Domain/Path based blocklists, Facebook / Klout integration.

Software Engineer, mobclix — Palo Alto, CA 01/11-03/12

- Designed and implemented algorithms for optimizing ad network allocations under each publisher, and for pacing the delivery of each ad campaign.
- Built the REST API third-party publishers / advertisers used to configure their ads on our exchange.
- Maintained our PHP based developer dashboard, and liaised with dev teams at third-party ad networks to help them integrate with our platform.

Software Development Engineer II, ***Microsoft*** — Mountain View, CA 05/07-05/10
Senior Software Engineer, Tellme. — Mountain View, CA 09/06-05/07

- Led prototyping Windows Phone 7 Speech Experience, and designed and developed UI/UX flows and speech grammars for the final WP7 Speech UX.
- Designed and Implemented a Voice + Visual application platform for Blackberry, Sprint Instinct, and other J2ME phones and led development of layout and rendering on Windows Mobile 6.5, and MobileML, a markup language spec for authoring apps.
- Built "Tellme for mobile" and "Transit" apps on top of MobileML.

Software Development Engineer, eBay — San Jose, CA 02/05-09/06
Senior Software Engineer, åccenture @ eBay — San Jose, CA 11/03-07/04

- As member of the K2 Search team, built a map-reduce based site layout analyzer on top of Nutch and Hadoop to detect and filter out redundant styling elements and other unimportant content. Also implemented heuristic site specific content scrapers using XPATH.
- As member of Stores team, also implemented features such as "Related Searches", and "Cross Seller Merchandizing" and moved the Stores codebase from C++/IIS based v2 architecture to Java/ WebSphere based v3 architecture.

Computer Scientist, Adobe (Flash Player team) — San Francisco, CA 05/10-01/11
Software Consultant, Spherion Consulting @ AOL — Dulles, VA 08/04-12/04
Java Developer, Tek Systems @ Genentech — Vacaville, CA 03/02-07/03
Software Development Intern, ***Microsoft*** — Redmond, WA 06/01-09/01
Software Development Intern, Cisco Systems — San Jose, CA 06/00-01/01

Education

University of California, Davis, CA — B.S. Computer Engineering, 2002

Coursera Specializations: Stanford MMDS (w/ distinction), Stanford ML, DeepLearning.ai DL